

Dataset Overview & Pre-Model Processing

This analysis leverages a comprehensive dataset of transaction-level hotel reservations to evaluate the causal impact of recommendation-driven bookings on high-value revenue generation. Rigorous preprocessing and feature engineering protocols were applied to isolate the true revenue-generating population. The data was strictly filtered to include only confirmed, payment-verified, and fully fulfilled bookings, eliminating noise from unmaterialized transactions. Financial features were standardized to Euros using transaction-specific exchange rates to neutralize currency volatility as an unobserved confounder. Furthermore, after engineering critical contextual features like city popularity indices, the "high-value" target variable (defined as the top 25% of net commission revenue) was computed exclusively on the training distribution.

Motivation

Hotel reservation systems generate large amounts of booking data. Understanding different customer segments can help hotels improve pricing, marketing strategies, and recommendation systems.

Since no predefined customer groups exist, an **unsupervised learning** approach was chosen to automatically discover hidden booking patterns.

Economic Relevance

The identified customer clusters can help hotels to:

Improve recommendation systems - Increase high-value bookings - Design targeted discount strategies - Better understand business (B2B) and leisure customers - Optimize hotel pricing and marketing campaigns.

Methodology

Step 1 – Feature Selection

Different feature combinations were evaluated to determine which variables best separate booking behaviors.

Because Gower distance requires an **N × N distance matrix**, a random sample of the dataset was used to reduce computational cost.

Gower distance was selected because it supports mixed numerical and categorical variables.

For numerical variables: $s_{ij} = \frac{|x_i - x_j|}{\text{Range}}$.

using normalized Manhattan distance.

For categorical variables: 0 = same category, 1 = different category .

The overall distance is: $D = \frac{P_n \cdot D_{num} + P_c \cdot D_{cat}}{P_n + P_c}$.

Parallel processing was applied to evaluate many feature combinations efficiently.

The best feature combination was: hotel_rank, discount, is_b2b, book_from_recommendation

With K = 4 and Silhouette Score = **0.947**. Other results were shown in [Table 3- 1](#).

Step 2 – label sample dataset and draw dendrogram diagram

Calculate distance matrix with Gower distance with selected features and then labeled sample dataset and then calculate quadratic distance to use **Average Linkage Hierarchical Clustering** method to hierarchically cluster objects from bottom to up which you can see that on [Figure 3- 1](#).

Step 3 – Identify medoids and label main data set

Find medoid of each cluster (the object which has minimum distance with other neighbors in its cluster) then label all the main data set by calculating distance from the medoids and then label object by closest medoid.

Step 4 – Cluster Assignment

Medoids (the most representative observation of each cluster) were identified.

Each remaining booking was assigned to the nearest medoid, allowing the complete dataset to be clustered.

Results

Cluster Characteristics

Cluster 0 – High-Value Discount Cluster

Largest cluster (~35,000 bookings) - Highest number of high-value reservations (~9,000) - Highest average discount (~13 €).

This suggests that discounts are positively associated with high-value bookings.

Cluster 1 – Luxury Hotels

Highest average hotel rating (~4 stars) - Very few recommendation bookings (~1%) - Almost all B2B reservations belong to this cluster.

Cluster 2 – Economy Hotels

Highest recommendation rate (~75%) - Average hotel rating around 2 stars - Represents the economy hotel segment.

Cluster 3 – Low-Ranking Hotels

Worst average hotel rank (~191) - Around 9% high-value bookings - About 3% recommendation bookings

Main Findings from the Figures

[Figure 3- 2](#): Most bookings belong to Cluster 0.

[Figure 3- 3](#): The recommendation system mainly recommends Cluster 2 (economy hotels), although relatively few high-value bookings occur there.

[Figure 3- 4](#): Cluster 1 clearly represents luxury hotels, while Cluster 2 contains economy hotels.

[Figure 3- 5](#): Cluster 3 has the poorest hotel rankings.

[Figure 3- 6](#): Higher discounts are associated with a higher proportion of high-value bookings.

[Figure 3- 7](#): Most B2B reservations occur in the luxury hotel cluster.

Limitations

- Gower distance requires an expensive $N \times N$ distance matrix, making it computationally intensive for large datasets.
- Random sampling was necessary to reduce memory usage.
- Cluster quality depends heavily on the selected features.
- Hierarchical clustering is slower than methods such as K-Means for very large datasets.

Synthesis

This study demonstrates that combining **Gower Distance** with **Hierarchical Clustering** effectively identifies meaningful customer segments in mixed-type hotel booking data.

The discovered clusters reveal clear differences in hotel quality, recommendation behavior, discounts, and business reservations.

The analysis also suggests that the current recommendation system mainly promotes economy hotels, while higher discounts appear to encourage high-value bookings.

These insights can support more effective pricing strategies, customer segmentation, and recommendation system improvements.

Support

Experiment	Feature Combination	Feature Count	K	Silhouette Score
1	lead_time_days, stay_nights, discount, final_price, is_b2b, book_from_recommendation	6	2	0.8317
2	lead_time_days, stay_nights, discount, final_price, is_b2b, book_from_recommendation	6	2	0.8317
3	hotel_rank, discount, is_b2b, book_from_recommendation	4	3	0.9317
4 (Best)	hotel_rank, discount, is_b2b, book_from_recommendation	4	4	0.9461
5	hotel_rank, discount, is_b2b, book_from_recommendation	4	4	0.9461
6	sum_room_extra_bed, hotel_rank, discount, is_b2b, book_from_recommendation	5	4	0.8795
7	All 14 features	14	2	0.2890
8	lead_time_days, stay_nights, sum_room_children, sum_room_extra_bed, hotel_rank, discount, final_price, is_b2b	8	2	0.7856
9	lead_time_days, stay_nights, discount, final_price, is_b2b, book_from_recommendation	6	2	0.8317

Table 3- 1

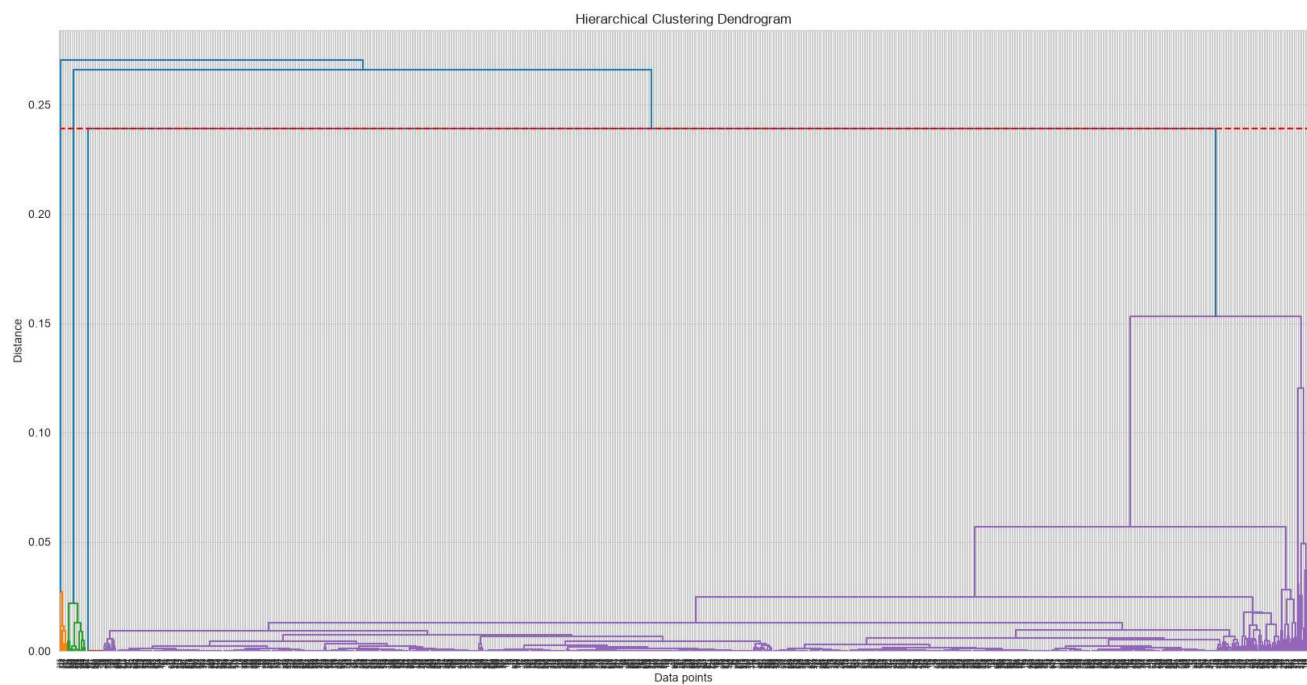


Figure 3- 1

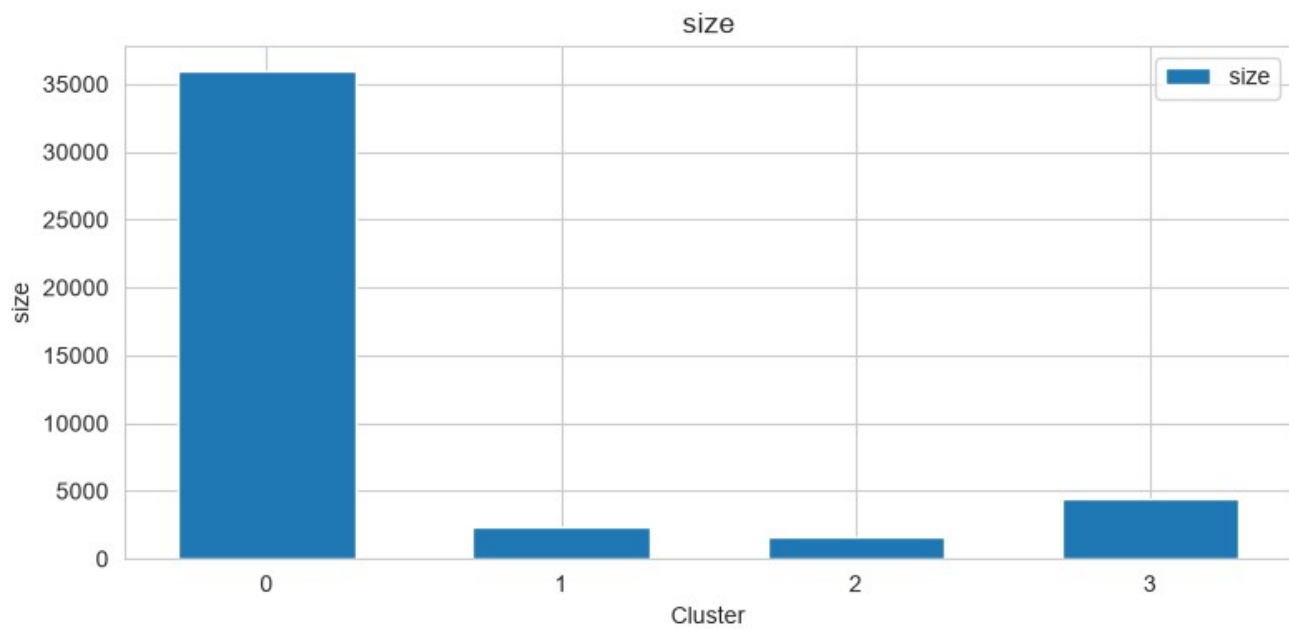


Figure 3- 2

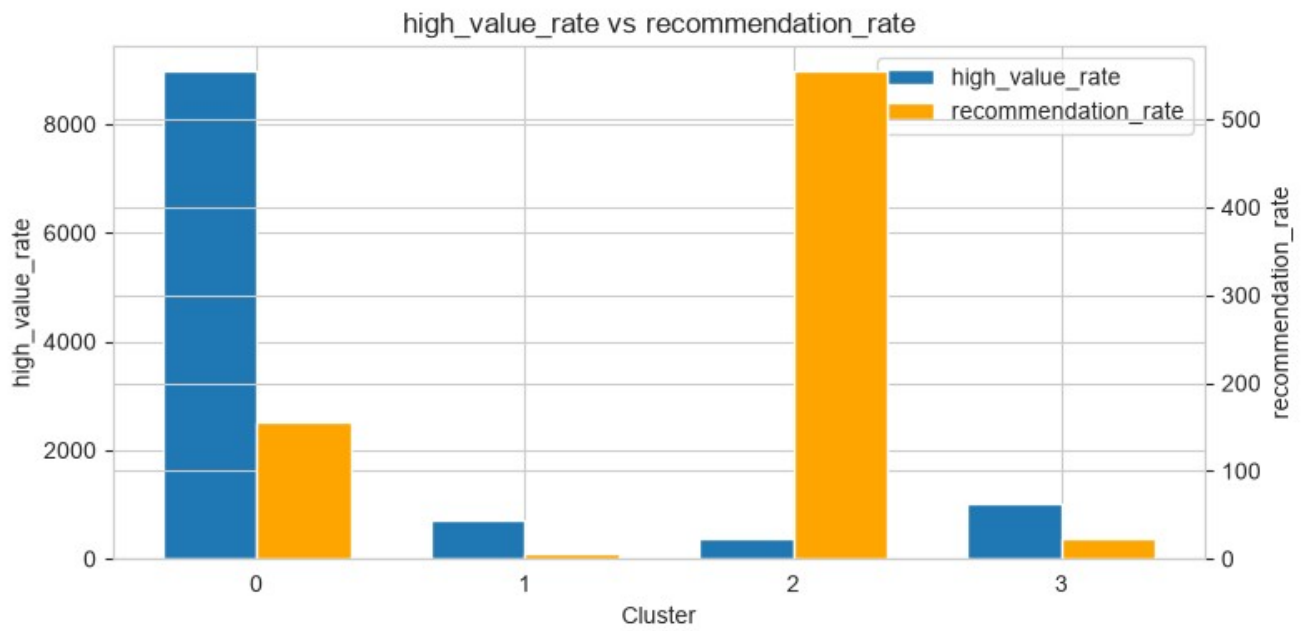


Figure 3- 3

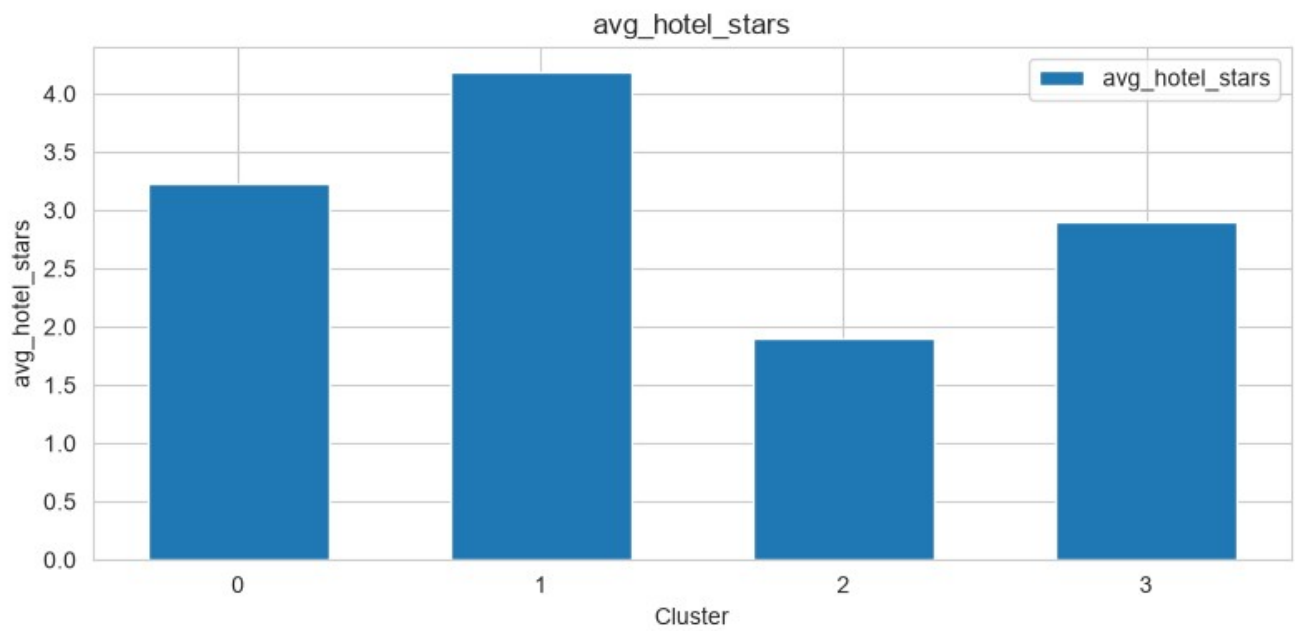


Figure 3- 4

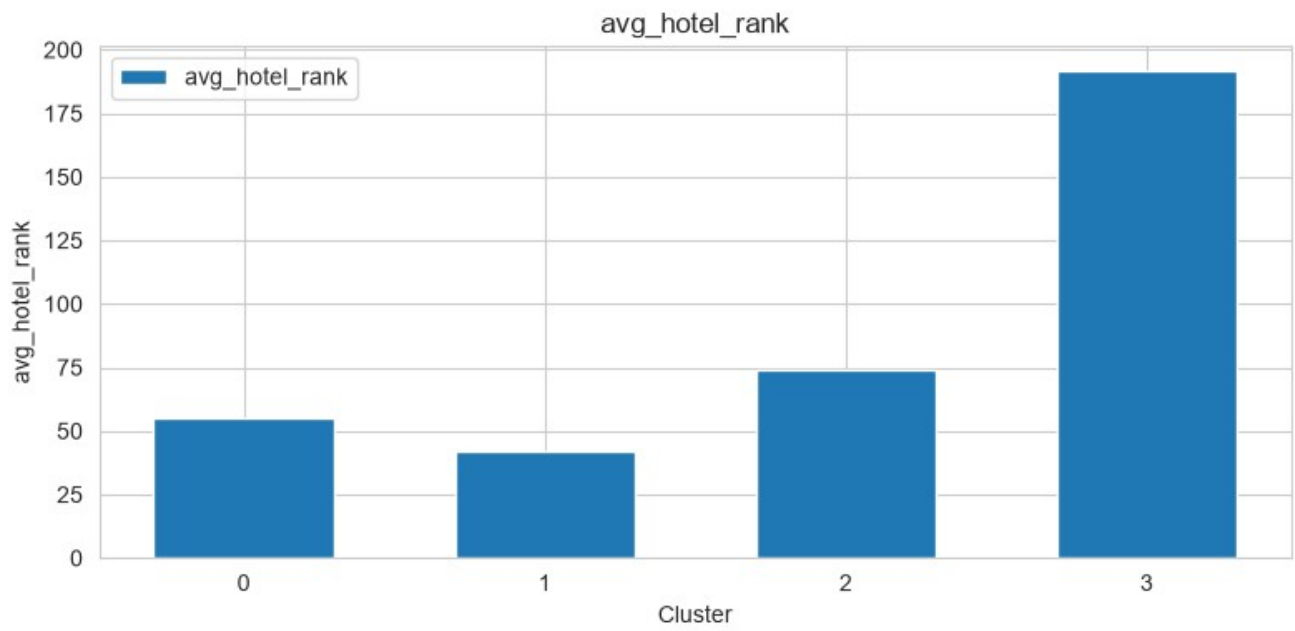


Figure 3- 5

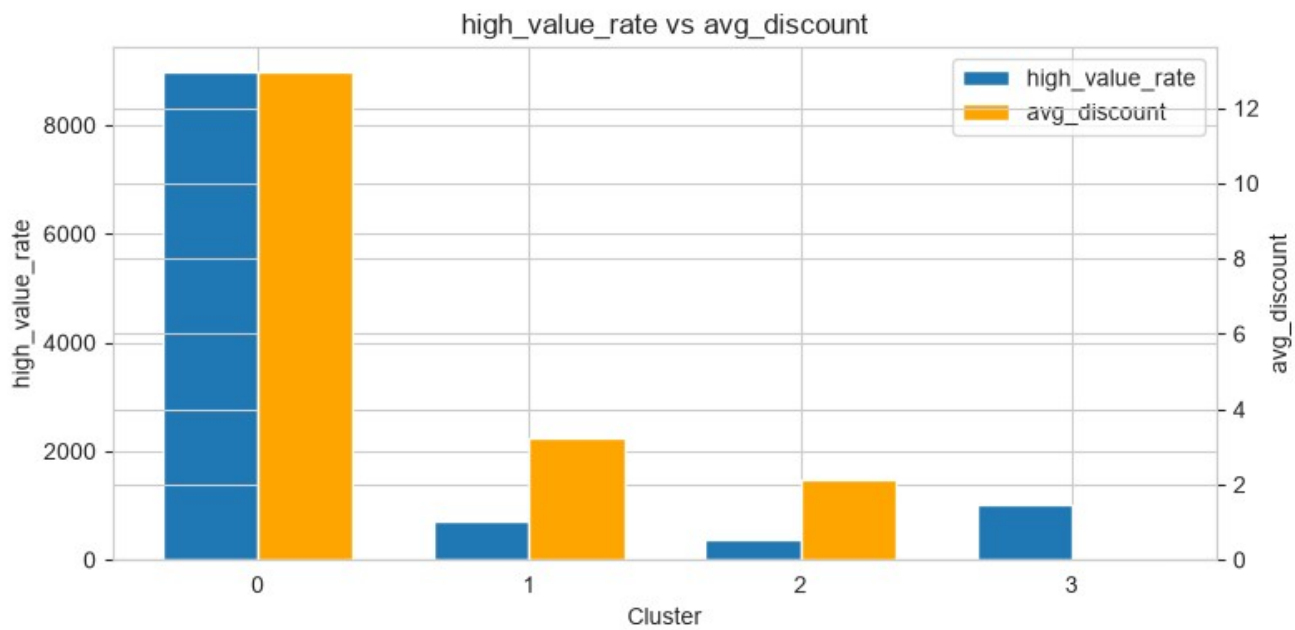


Figure 3- 6

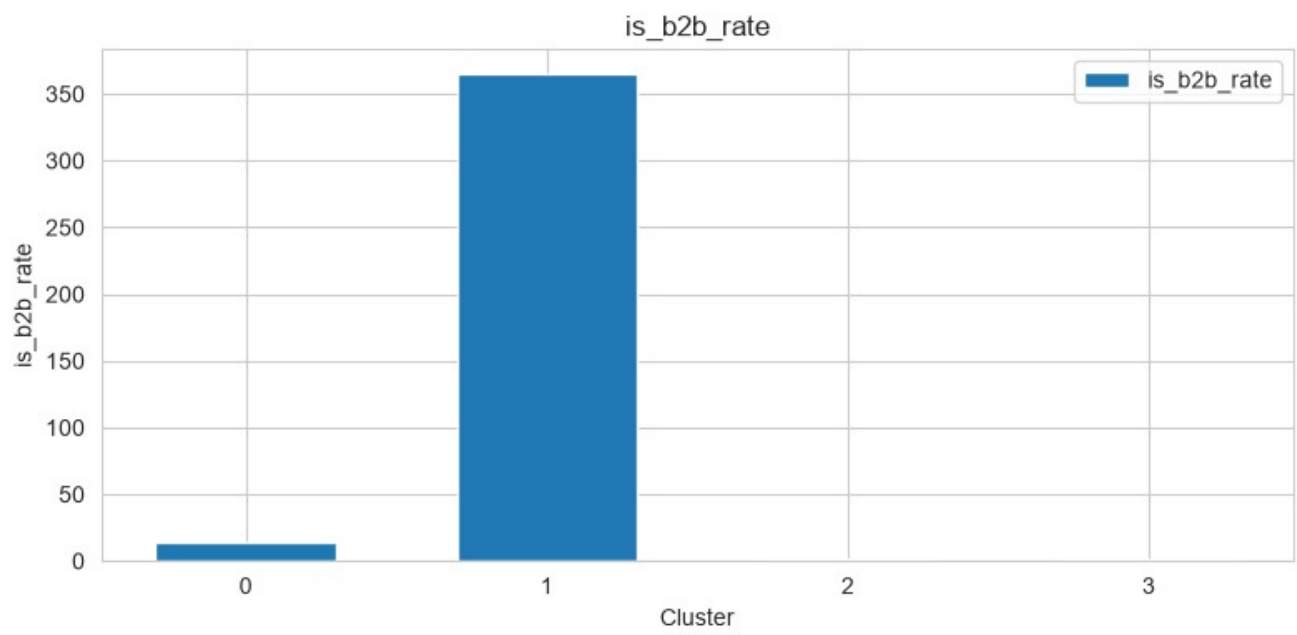


Figure 3- 7